Long-Term Agent Memory

Design, implementation and evidence-based evaluation

5 November 2026

Abstract. Six memory prototypes expose different ways to retain or lose conversational evidence. This study traces failures through storage, retrieval, answer generation and evaluation. Source-preserving retrieval repairs yield 20/28 correct answers for Graph RAG versus 18/28 for History RAG in a small diagnostic study. Additional baselines, runtime measurements and evaluator checks extend the comparison. The contribution is a reproducible account of design decisions and failure boundaries, not a claim of benchmark superiority.

1 Problem description

Long-term agent memory is persistent state selected from past interactions and reused in later tasks. A useful memory must preserve who asserted a fact, when it was valid and which source supports it. A log stores events; a conversation window stores recent text. Neither alone decides what to retain, update, retrieve or forget.

Example 1. An illustrative update and provenance problem.

Time / speaker	Interaction
t1 / user	My office is in Seoul.
t2 / assistant	You could relocate to Busan.
t3 / user	I moved my office to Daejeon.
query	Where is my office now? → Daejeon

Finite windows evict old facts. Summaries and schemas may discard a title, quantity or date before the future question is known. Retrieval can miss one operand of a multi-session answer; the reader can still select an obsolete value. The experiments therefore inspect evidence survival at each stage, following LongMemEval's indexing-retrieval-reading decomposition [2].

2 Research basis and scope

Table 1. Research ideas and the implemented scope.

Research	Mechanism	Scope here
GraphRAG [1]	Entity/relation extraction, communities and global summarization.	Local property-graph retrieval with source evidence; no community summaries or global map-reduce.
LongMemEval [2]	Long histories, updates, multi-session reasoning and abstention.	Cleaned S data, structural audit, diagnostic samples and semantic grading.
HippoRAG 2 [3]	Passage-integrated graph retrieval and personalized PageRank.	Conceptual reference only; no implementation or head-to-head score.

3 Memory designs: retaining the conversation

The code images excerpt the actual notebook; ellipses omit unrelated arguments or blocks. Outputs below are saved observations, not synthetic examples. All answer models use qwen3:8b; the summarizer uses qwen3:1.7b.

3.1 No Memory

```
messages = [
    system_prompt,
    HumanMessage(question),
]

answer = chat.invoke(
    input=messages,
    config=config,
)
```

```
Q: What degree did I graduate with?
Output: Bachelor's degree in Computer Science.
Reference: Business Administration.
```

Observation. The saved notebook demo invents a private fact. No ingestion or retrieval is required, but a question-only model cannot recover information absent from its input. Abstention must be measured separately from successful recall.

3.2 Chat History

```
messages = []
for message in reversed(flatten):
    if count_tokens_approximately([message, *messages]) < max_history_length:
        messages.insert(0, message)
    else:
        break
```

```
Q: What degree did I graduate with?
Output: I don't know.
Retained context: two kitchen-pantry messages.
```

Observation. A 1,000-token window preserves wording but loses the degree evidence. The implementation takes the last messages in dataset order, which is not always chronological. Its strength is simple bounded context, not long-term retention.

3.3 Summarization

```
summarizer = SummarizationMiddleware(
    model=ChatOllama(model="qwen3:1.7b", num_predict=384, ...),
    trigger=("tokens", 6_000),
    keep=("tokens", 512),
    ...
)
...
result = summarizer.before_model({"messages": messages}, runtime)
if result:
    messages = result["messages"][1:]
```

```
Q: What play did I attend at the local community theater?
Output (excerpt): The conversation does not mention any play attended ...
Reference: The Glass Menagerie. (24 compressions; 25.72 s)
```

Trade-off. Compression at 6,000 tokens retains the last 512 tokens and asks for a running summary of at most 150 words. It extends continuity, but repeated LLM calls add ingestion cost and exact facts can disappear before the query arrives. The supplementary run replays the unchanged middleware.

3 Memory designs: retrieval and structure

3.4 History RAG

```
dense = store.similarity_search(
    "Represent this sentence for searching relevant passages: " + question, k=5
)
sparse = bm25.get_top_n(tokenize(question), chunks, n=5)
candidates = list(dict.fromkeys([doc.page_content for doc in dense] + sparse))

ranked = reranker.rank(question, candidates, top_k=3)
retrieved = [candidates[hit["corpus_id"]] for hit in ranked]
```

```
Q: What play did I attend at the local community theater?
Output: The play you attended was "The Glass Menagerie."
Reference: The Glass Menagerie. [saved 28-item experiment]
```

Observation. Dense top-5 plus BM25 top-5 candidates and cross-encoder top-3 selection recover an exact title. Indexing needs embeddings but no extraction LLM. Raw chunks remain interpretable, yet there is no consolidation/forgetting policy and a small retrieval budget can miss part of an answer.

3.5 Structured Memory

```
class Episode(BaseModel):
    who: str = Field(description="Who performed the action.")
    action: str = Field(description="Short action or event.")
    whom: str = Field(...)
    when: str = Field(...)
    where: str = Field(...)
```

```
Representation: Episode(who, action, whom, when, where)
Persistence: SQLite; agent-accessible search.
Inference output / benchmark score: not measured.
```

Scope. Working, semantic, episodic and procedural memory were studied conceptually; this prototype implements episodic records. Typed fields improve inspection, but per-message extraction is expensive and the schema may omit future-useful details. Deletion is disabled. It is excluded from the new benchmark for cost, not assigned an inferred score.

3.6 Graph RAG

```
selected = {}
ranked = (
    reranker.rank(question, [text for _, text in evidence], top_k=min(20, len(evidence)))
    if evidence else []
)
for hit in ranked:
    source_id, excerpt = evidence[hit["corpus_id"]]
    if source_id not in selected:
        if len(selected) == 10:
            continue
        selected[source_id] = []
    selected[source_id].append(excerpt)
```

```
Q: How old was I when my grandma gave me the silver necklace?
Evidence: "... on my 18th birthday."
Output: The user was 18 years old ... [fresh two-item run]
```

Observation. Entity relations support associative access, while original dialogue retains details lost in extraction. Global reranking selects up to 20 excerpts across 10 source blocks. This local variant remains costly to index and can misread competing facts even when source evidence is present.

3.7 Knowledge graph visualization

The exported views inspect the same single-sample graph used to answer the question below. The full view contains all stored nodes and relations. The retrieved view retains exactly the relation hints included in the final answer context; text retrieved through dense/BM25 search is a separate evidence channel.

```
Question: What degree did I graduate with?
Answer: You graduated with a degree in Business Administration.
```

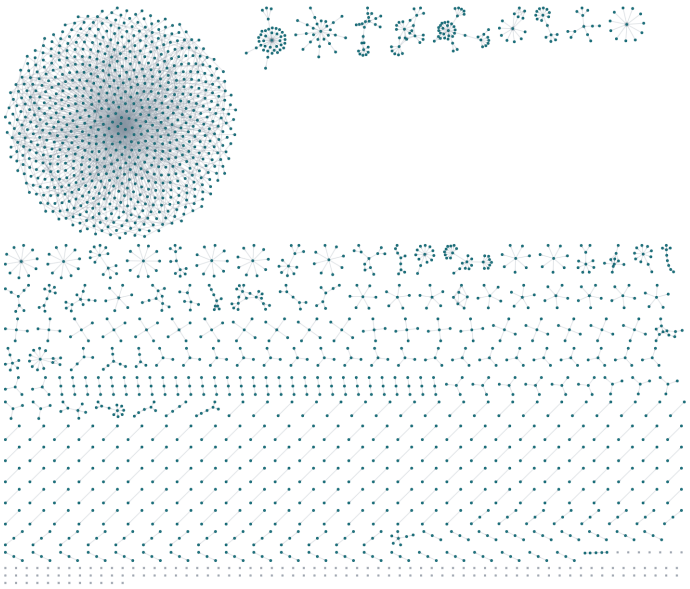


Figure 1. Full graph: 2,184 entity nodes, 147 source chunks and 1,937 relations. Source chunks are gray squares; entities are teal. HTML supports zoom and hover inspection.

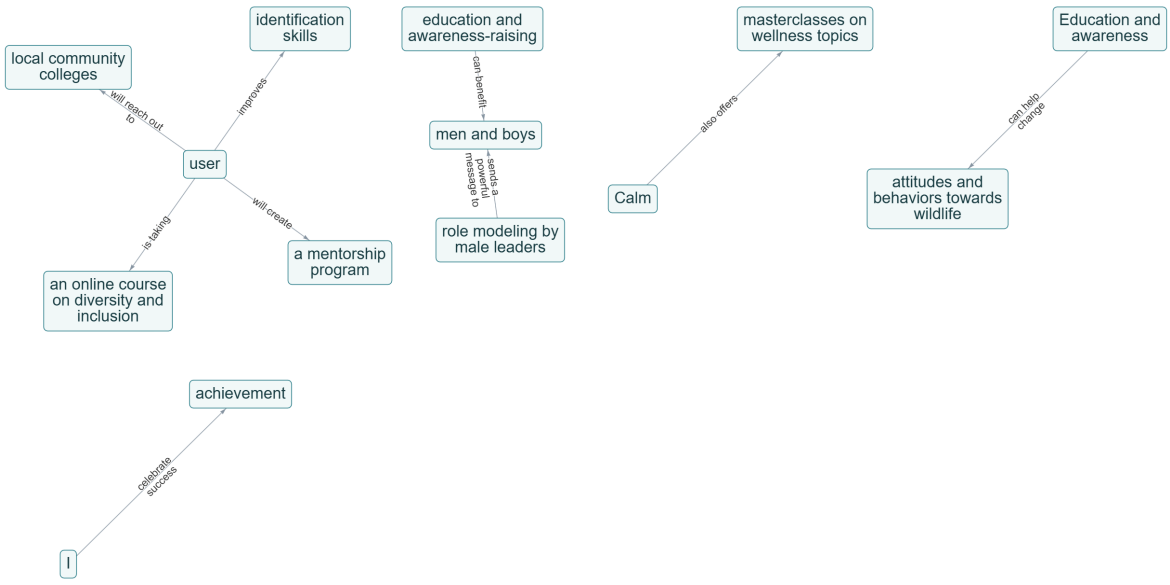


Figure 2. Retrieved subgraph: 14 nodes and 9 exact relations. Multiple predicates between the same directed node pair share one drawn edge; all are retained in its tooltip.

The visualization error came from parsing a “Conversation excerpts” block as a triple. The fix extracts only explicit “Graph relations” and creates no edges for text-only evidence. The degree entity exists in the full graph but is absent from the selected hints; retrieved dialogue supplies the answer. This distinguishes graph retrieval from source-evidence recovery.

4 Failure investigation and trial decisions

A stronger answer model alone does not repair evidence lost upstream. Diagnosis followed the path from raw session to relation, retrieved source, selected excerpt, generated answer and grade. Labels and references were used for analysis only; retrieval never filters history by the gold answer.

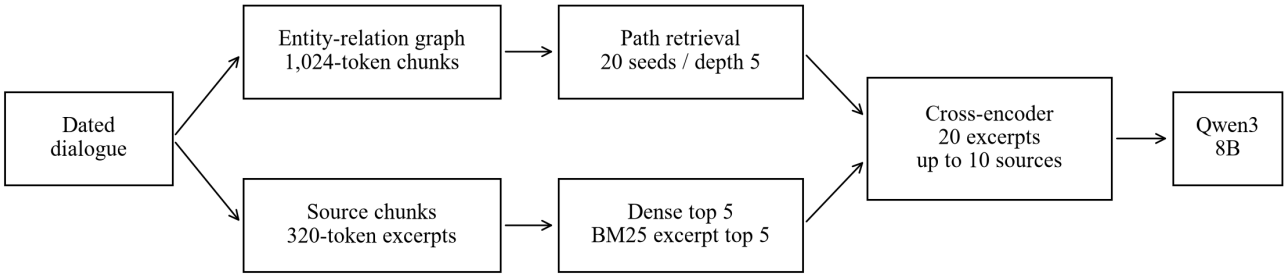


Figure 3. Final source-preserving Graph RAG pipeline. Dialogue precedes optional relation hints in the answer context.

Table 2. Diagnostic trials; these are not a controlled aggregate ablation curve.

Attempt	Observed evidence	Decision
160 → 320-token excerpts	Case 58bf7951 still omitted the play title when only two excerpts per source survived.	Size alone was insufficient.
Global reranking	Same cached graph, prompt and 320-token splitter: title recovered; answer became correct.	Keep bounded global selection.
Fine source search	75832dbd: coarse BM25 source rank 49/137; fine excerpt rank 5. Header recovery passed 346/346 checks.	Add dense sources + fine BM25.
Shared prompt rewrite	Development: History 11 vs 12 original; Graph 12 + 1 ambiguous vs 13 original.	Reject; retain original prompt.

Table 3. Final settings; unequal evidence budgets remain a limitation.

Component	Configuration
Answer model	qwen3:8b; reasoning on; temperature 0.6; top_p 0.95; top_k 20; context 32,768; output 2,048; seed 42
Extractor	qwen3:4b-instruct; temperature 0; no thinking; requests 10 paths/chunk; one worker.
Encoder / reranker	BGE-small-en-v1.5 / ms-marco-MiniLM-L6-v2
Graph retrieval	1,024-token sources; 320-token excerpts; overlap 32. 20 entity seeds; depth 5; at most 512 path candidates.
Context budget	History: top 3 excerpts. Graph: up to 20 excerpts / 10 source blocks, subject to a final token guard.

5 Benchmark and experimental protocol

LongMemEval is an established long-term conversational-memory benchmark introduced at ICLR 2025 [2]. It is also used in industrial memory evaluations, including Zep and Mem0 [7,8]. This adoption makes it a useful shared testbed; it is not a formal industry standard, and vendor scores under different settings are not directly comparable.

Its questions test personal facts, earlier assistant responses, preferences, cross-session reasoning, changing knowledge and unanswerability. The difficulty is not merely a long input: evidence can be indirect, scattered across sessions or contradicted by later facts. Background dialogue creates distractors, and plausible advice may still miss the intended preference.

Table 4. Dataset specification. Paper settings and local audited statistics are separated.

Item	Specification
Canonical benchmark [2]	500 curated questions; timestamped multi-session user-assistant histories. S ≈115k tokens/history; M ≈1.5M. Oracle retains evidence sessions only.
Construction [2]	Human-curated questions; simulated, human-edited evidence dialogues; background includes simulated sessions, ShareGPT and UltraChat.
This dataset [6]	xiaowu0162/longmemeval-cleaned / longmemeval_s_cleaned.json
Cleaned release	September 2025 cleanup removes interfering noisy sessions; the original paper predates this version.
Local audit	500 rows; 38-62 sessions/history; median 48. Paper token counts are not remeasured cleaned-file counts.
Available fields	question / answer / question_date / haystack_sessions / haystack_dates / answer_session_ids; turn-level has_answer annotation
Audit findings	211/500 histories unsorted; 41 answerable cases have same-day gold sessions after the question clock time. A strict timestamp cutoff can remove valid evidence.

Table 5. Mutually exclusive effective types and sample allocation.

Type	Cleaned set	Development	Validation
Single-session user	64	2	2
Single-session assistant	56	2	2
Preference	30	2	2
Multi-session	121	2	2
Temporal reasoning	127	2	2
Knowledge update	72	2	2
Abstention	30	2	2
Total	500	14	14

IDs ending in `_abs` override the six stored question-type labels, giving seven effective types. Selection uses seed 42 and four examples per type. The original development split informed retrieval choices; validation was separate. New baselines use the same frozen selection. A balanced diagnostic sample is not a prevalence-weighted estimate of the 500-question score.

6 LLM-as-a-judge: specification and validation

The official evaluator uses question, reference/rubric and candidate response, without the full history. Its example configuration is GPT-4o-2024-08-06, temperature 0 and a yes/no answer [9]. The local implementation substitutes qwen3:8b and parses an ordinary JSON boolean. It adapts the rubric; it does not reproduce the official evaluator or establish model equivalence.

Table 6. Official semantic rules and the local adaptation [9].

Case	Official rule (paraphrased)	Local treatment / caveat
Fact / multi-session	Equivalent wording or all necessary intermediate facts; incomplete subsets fail.	Same intent; an extra “estimated to know” clause is more permissive.
Time / update	Numeric duration ±1 allowed; old information may coexist with the required new value.	Type-specific rules distinguish durations, calendar dates and current/past values.
Preference	Partial rubric coverage can suffice if personal information is correctly recalled and used.	Local wording requires the reference preference more explicitly; potentially narrower.
Abstention	Recognize that the requested information is unavailable.	Require no definite guess; synthetic controls gate execution.

```
class Grade(BaseModel):
    correct: bool = Field(strict=True)
...
judge = chat.model_copy(update={"temperature": 0})
answer = judge.invoke(...)  # ordinary JSON; no forced schema
return Grade.model_validate_json(answer.content)
```

Listing 1. Abbreviated local implementation. Strict parsing checks output type, not semantic correctness.

Table 7. Evaluator validation and remaining failures.

Probe	Evidence	Action / limit
Forced JSON Schema	Same real refusal at T=0: false, true, true despite negative reasoning.	Remove forced schema; plain JSON repeats false, false, false.
Controls	24/24 extended controls; notebook gate 6/6.	Useful guard, insufficient for natural-answer reliability.
Fresh baseline answers	“Cannot be determined” graded true for a two-month reference; another refusal graded true for $400k. One empty JSON output was logged.	Retain raw judgments; use two blinded reviews for reported scores.

Reviewers see only type, question, reference and answer, with method names and automatic labels hidden. Disagreement is reported as an interval. These are independent model reviews, not human ground truth. The evidence establishes format and semantic failure modes; model size alone is not a proven cause.

7 Experimental results

The earlier quantitative work targeted the suspected Graph RAG regression, so the first 28-question comparison included only the two retrieval methods. That scope did not test the simpler baselines. The extension below addresses this gap while retaining the original notebook logic and the frozen RAG predictions.

Table 8. Correct answers on the same 28 questions. Ranges denote grading ambiguity, not confidence intervals.

Method / run	Development	Validation	Combined
No Memory (new)	1/14	1-2/14	2-3/28
Chat History, 1k (new)	2/14	2/14	4/28
Summarization (new)	3/14	3/14	6/28
History RAG (frozen)	12/14	6/14	18/28
Original Graph (frozen)	9/14	7-8/14	16-17/28
Fixed Graph (frozen)	13/14	7/14	20/28
Structured Memory	Not run	Not run	Not measured

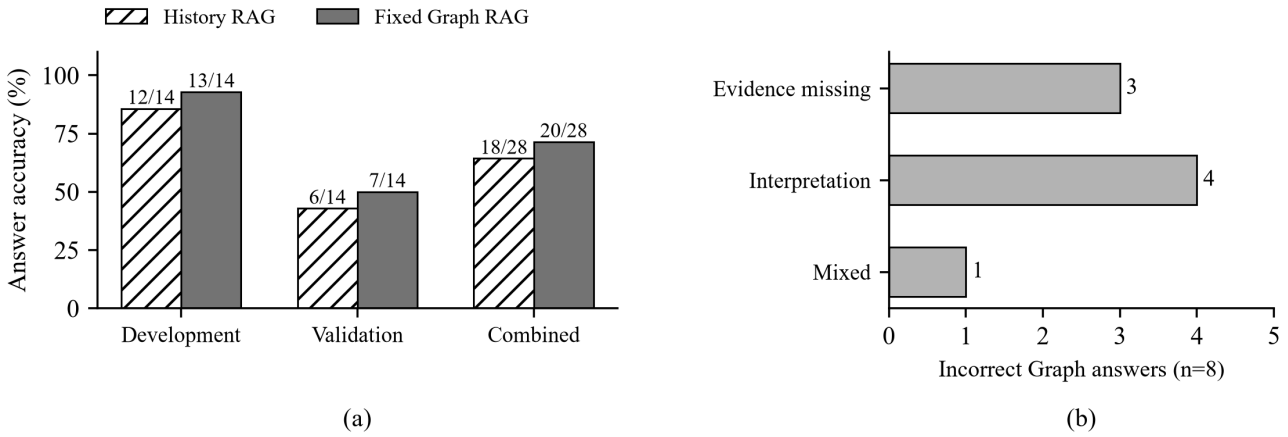


Figure 4. (a) Frozen RAG accuracy, with counts. (b) Diagnosis of eight fixed-Graph failures in that experiment. These errors exclude the separate two-item recommendation case.

History and fixed Graph were both correct on 17 items; Graph alone on 3; History alone on 1; both failed on 7. The net difference is two items. Different context budgets, development-based selection and small n prevent a general or statistical superiority claim.

Chat History's 4 correct answers are all abstentions; this is not successful personal-fact recall. Summarization reaches 6/28 despite 661 compression calls; 334 calls (50.5%) hit the 384-token output cap. Its temperature/seed remain unspecified notebook defaults. No Memory has one ambiguous location answer (closet without shoe rack). New results are single-run diagnostics; automatic labels are not the reported scores.

Table 9. Separate fresh-kernel demonstration, 5 November 2026.

Case	Reference / observed output	Saved grade
Viewing recommendation	Netflix storytelling stand-up / Egypt documentary	False*
Necklace gift age	18 / 18	True

The earlier 15-cell run completed without runtime errors, passed 6/6 controls and scored 1/2. Its 325.6 s includes setup and demonstrations. Section 3.7 is a separate later visualization run. *Section 8 distinguishes the saved label from practical utility.

8 Error analysis and computational cost

8.1 A reference miss can still be grounded in memory

Case 32260d93 asks for something to watch tonight. The reference favors Netflix stand-up with strong storytelling; the answer recommends a BBC ancient-Egypt documentary. The earlier report described this only as an interpretation failure. Inspecting both user statements supports a more precise diagnosis.

Table 10. Two genuine interests coexist in the retrieved history (paraphrased).

Source	Evidence	Implication
User / May 29, 04:39	Aspiring comedian asks for Netflix specials with storytelling like John Mulaney.	Supports the reference; more recent interest.
User / May 27, 16:48	Explicit interest in ancient Egypt; asks for documentaries or online resources.	Supports the alternative topic.
Retrieved relation	Earlier assistant recommendation of the same BBC title.	Explains the selected title; does not verify its existence.

The saved local label remains false. However, the Egypt topic is history-grounded and potentially useful, rather than unsupported. The official preference rubric allows partial coverage and correct use of personal information [9], but its grader receives no history and cannot verify this alternative interest. The reference says other genres “may” be disfavored, not that they are forbidden. This is a rubric-coverage and preference-selection ambiguity for history-aware human review; the official outcome has not been rerun, and the benchmark score is not silently changed.

8.2 Runtime: measure ingestion separately from answering

Table 11. Wall time in seconds. New runs are sequential; judging is excluded.

Method / sample	Memory build	Answer/query	Total
No Memory, n=2	<0.01	1.083	1.083
Chat History, n=2	<0.01	1.721	1.726
Summarization, n=2	25.187	2.085	27.310
History RAG, n=2	22.285	2.325	24.609
Graph RAG, prior n=2	Not split	Not split	141.643

Means on the same two regression items (137, 27); new runs use one RTX 5090. Memory build is replay/compression for Summary and fresh indexing for History RAG. The latter spends 22.285 s indexing versus 2.325 s retrieving/answering, so cost depends strongly on index reuse. Setup, controls and grading are excluded; controls warm the 8B model, and later in-call loads remain included. Two items do not characterize latency variance.

The original No Memory/Chat History timer accumulates across a list, so new measurements call each unchanged function with one item and time externally. Historical RAG timings mixed cached evidence and concurrent runs; they are not comparable end-to-end latency. The old two-item Graph run averaged 141.64 s including graph construction, a separate context rather than a controlled speed ratio.

9 Discussion and reproducibility

Table 12. What the experiments changed in the design reasoning.

Finding	Design implication
No memory invents facts; windows lose them.	Measure grounded recall and abstention, not only fluent answers. Preserve data order assumptions explicitly.
Compression and structure are lossy.	Retain retrievable original evidence; evaluate whether the needed detail survives before blaming the reader.
Evidence present does not ensure correctness.	Separate missing operands, stale-value selection and preference ambiguity. Report retrieval recall alongside answer accuracy.
Parseable grades can still be wrong.	Keep controls, raw model output and independent reviews. Distinguish benchmark compliance from practical utility.

Restoring embedding batches was also workload-dependent: async batch-10 versus sync batch-32 took 1.485 → 0.499 s for 128 short texts, but 9.098 → 11.135 s for long texts. Maximum embedding differences were 1.49e-7 and 0. These measurements establish numerical consistency, not an overall pipeline speedup.

Next evaluation should use all 500 questions, equal context-token budgets, multiple seeds and independent history-aware adjudication. Cold indexing and query latency need separate, amortized measurements. Consolidation, deletion, multi-agent sharing and production operation remain unestablished.

Future work. Test HeLa-Mem [4] for episodic associations and semantic consolidation under fixed reader/context budgets. It remains unimplemented.

Artifacts. The original RAG evidence is frozen at adebbe3 [5]. Supplemental predictions, timings and blinded reviews are stored under experiments/baseline_extension_20261105. The runner imports current notebook definitions; it does not modify memory logic. New stochastic generations need not reproduce frozen predictions exactly.

```
python experiments/compare_baselines.py --help
python experiments/reproduce.py --split all --run
```

References

[1] D. Edge et al. *From Local to Global: A GraphRAG Approach to Query-Focused Summarization.* 2024, revised 2025. arXiv:2404.16130.

[2] D. Wu et al. *LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory.* ICLR 2025. arXiv:2410.10813.

[3] B. Jimenez Gutierrez et al. *From RAG to Memory: Non-Parametric Continual Learning for Large Language Models.* ICML 2025. arXiv:2502.14802.

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[5] aruwad. *ai-topic-memory.* Code, frozen RAG results and reproduction scripts. Project repository, adebbe3.

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[8] Mem0. *Memory Evaluation.* Official project documentation. Evaluation documentation.

[9] LongMemEval. *evaluate_qa.py.* Official grading implementation, d6dc8b5. Source and type-specific rubrics.